

Time dependent Schrodinger Equation

$$\Psi = A e^{i(kx - \omega t)} \rightarrow (1)$$

$$k = \frac{2\pi}{\lambda} ; \omega = 2\pi\nu ; E = h\nu$$

$$\lambda = \frac{h}{p} ; \omega = 2\pi \times \frac{E}{h}$$

$$k = \frac{2\pi}{h} \times p$$

$$\omega = \frac{E}{\hbar} ; \cancel{E = \hbar \omega}$$

$$k = \frac{p}{\hbar}$$

$$\cancel{p = \hbar k}$$

$$\Psi = A e^{i(kx - \omega t)}$$

$$\Psi = A e^{i\hbar(p x - E t)}$$

$$\frac{\partial \Psi}{\partial t} = A e^{i\hbar(p x - E t)} \cdot \left(-\frac{i}{\hbar} E\right)$$

$$\frac{\partial \Psi}{\partial t} = \left(-\frac{i}{\hbar} E\right) \Psi \Rightarrow i\hbar \frac{\partial \Psi}{\partial t} = E \Psi \rightarrow (2)$$

$$\frac{\partial \Psi}{\partial x} = A e^{i\hbar(p x - E t)} \cdot \frac{i}{\hbar} p$$

$$\frac{\partial^2 \Psi}{\partial x^2} = A e^{i\hbar(p x - E t)} \cdot \frac{i^2}{\hbar^2} p^2$$

$$\frac{\partial^2 \Psi}{\partial x^2} = -\frac{p^2}{\hbar^2} \Psi \Rightarrow p^2 \Psi = -\hbar^2 \frac{\partial^2 \Psi}{\partial x^2}$$

The total energy of the particle is.

$$E = \frac{p^2}{2m} + V$$

$\downarrow$ $\downarrow$
 KE PE

$$E\psi = \frac{p^2}{2m}\psi + V\psi$$

$$\underline{\underline{i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi \longrightarrow (2)}}$$

In 3D.

$$\underline{\underline{i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi}}$$

$\downarrow$
time dependent Schrodinger equation.

Time independent Schrodinger Equation.

$$\psi(x,t) = \psi(x)\phi(t) \longrightarrow (5)$$

$$\frac{\partial \psi}{\partial t} = \psi(x) \frac{\partial \phi(t)}{\partial t} \longrightarrow (3)$$

$$\frac{\partial^2 \psi}{\partial x^2} = \phi(t) \frac{\partial^2 \psi}{\partial x^2} \longrightarrow (4)$$

Putting (3) & (4) in (2)

$$i\hbar \psi(x) \frac{\partial \phi}{\partial t} = -\frac{\hbar^2}{2m} \phi(t) \frac{\partial^2 \psi}{\partial x^2} + V \psi(x) \phi(t)$$

Dividing throughout by $\psi(x) \phi(t)$

$$\frac{i\hbar}{\phi(t)} \frac{\partial \phi}{\partial t} = -\frac{\hbar^2}{2m} \frac{1}{\psi(x)} \frac{\partial^2 \psi}{\partial x^2} + V \rightarrow (8)$$

L.H.S is a function of t .

R.H.S is a function of x .

Hence, both must be equal to same constant K .

$$\frac{i\hbar}{\phi(t)} \frac{\partial \phi}{\partial t} = K$$

$$\frac{\partial \phi}{\phi} = \frac{1}{i\hbar} K dt$$

$$\frac{\partial \phi}{\phi} = \frac{-i}{\hbar} K dt$$

On integrating.

$$\int \frac{\partial \phi}{\phi} = \int \frac{-i}{\hbar} K dt$$

$$\log \phi = \frac{-i}{\hbar} K t$$

$$\phi = e^{-\frac{i}{\hbar} K t}$$

$$\phi = e^{-\frac{i}{\hbar} K t}$$

$$\psi(x, t) \text{ becomes } \Rightarrow \psi(x, t) = \psi(x) e^{-\frac{i}{\hbar} K t}$$

$$\frac{\partial \psi}{\partial t} = -\frac{iK}{\hbar} \psi(x) e^{\frac{-iKt}{\hbar}}$$

$$\frac{\partial \psi}{\partial t} = -\frac{i}{\hbar} K \psi$$

$$\underline{i\hbar \frac{\partial \psi}{\partial t} = K\psi \rightarrow (6)}$$

Comparing (6) & (7)

K is identical with E .

Equating the RHS of eqn (8).

$$-\frac{\hbar^2}{2m} \frac{1}{\psi(x)} \frac{\partial^2 \psi}{\partial x^2} + V = E$$

Multiplying by $\psi(x)$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi = E\psi$$

$$\frac{\partial^2 \psi}{\partial x^2} - \frac{2m}{\hbar^2} V\psi + \frac{2m}{\hbar^2} E\psi = 0$$

$$\underline{\frac{\partial^2 \psi}{\partial x^2} + \frac{2m}{\hbar^2} (E - V)\psi = 0} \rightarrow \text{1D.}$$

$$\underline{\nabla^2 \psi + \frac{2m}{\hbar^2} (E - V)\psi = 0} \rightarrow \text{3D.}$$